SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING

SUB CODE & NAME: CSA17- ARTIFICIAL INTELLIGENCE

Exp 3: Write the python program for Water Jug Problem

CODE:

```
from collections import deque

CAP_A, CAP_B = 4, 3
START = (0, 0)
GOAL = 2

def solve_water_jug():
    queue = deque([(START, [START])])
    visited = {START}

    while queue:
        (a, b), path = queue.popleft()

        if a == GOAL or b == GOAL:
            return path

        transfer = min(a, CAP_B - b)
        transfer2 = min(b, CAP_A - a)

        next_states = [
            (CAP_A, b),              # Fill A
            (a, CAP_B),              # Fill B
```

```
            (0, b),                 # Empty A
            (a, 0),                 # Empty B
            (a - transfer, b + transfer),
            (a + transfer2, b - transfer2)
        ]

        for state in next_states:
            if state not in visited:
                visited.add(state)
                queue.append((state, path + [state]))

    return None

path = solve_water_jug()

if path:
    for state in path:
        print(state)
else:
    print("Solution not found")
```

OUTPUT:

